Inductor Design Using E-E Cores: Area-Product Method

Introduction

This document sets out the general procedure for designing an inductor on an E-E core using the area-product method. It covers four steps: choosing the core from the catalog, finding the number of turns, finding the air gap length, and sizing the wire.

Design specifications

The design starts from six quantities, which come from the converter design and from the choice of material and winding limits.

Symbol	Quantity	Unit used in the formulas
L	Required inductance	H
I_p	Peak inductor current	A
I_{rms}	RMS inductor current	A
I_{avg}	Average inductor current	A
J	Current density in the winding	A/mm²
B_m	Maximum flux density	T (Wb/m²)
K_w	Window utilization (fill) factor	dimensionless

Step 1: Core selection

The core is chosen so that the product of its core area A_c and window area A_w is at least the required area product. Here A_c is the core cross-sectional area and A_w is the window area available for the winding.

The window must hold N turns of wire carrying I_{rms} at current density J, so the window area needed is:

$$A_w = N \cdot I_{rms} / J \qquad \dots (1)$$

The peak flux linkage equals $L \cdot I_p$, and the core carries it at flux density B_m:

$$N \phi_p = L I_p \;\Rightarrow\; N A_c B_m = L I_p \qquad \dots (2)$$

Multiplying (1) and (2) cancels N and gives the area product. The window utilization factor K_w is included so the winding does not assume the whole window is copper:

$$A_c \times A_w = (I_{rms} \cdot I_p \cdot L) / (B_m \cdot J \cdot K_w) \quad cm^4$$

Select the E-E core from the catalog whose $A_c \times A_w$ is equal to or just above this value.

Step 2: Number of turns

The turns follow from equation (2), using the A_c of the core selected in Step 1:

$$L I_p = N B_m A_c \;\Rightarrow\; N = L I_p / (B_m A_c)$$

Here B_m must be in Wb/m² (tesla) and A_c in m². Round N up to a whole number of turns.

Step 3: Air gap length

The inductor must hold L with the N turns found in Step 2, so an air gap is needed in the magnetic path. Neglecting the core reluctance and fringing flux, the gap length is:

$$l_g \approx \mu_0 N^2 A_c / L$$

Here $\mu_0 = 4\pi \times 10^{-7}$ H/m, N is the rounded number of turns, A_c is in m² and L is in H, so l_g comes out in metres. This is the total gap in the magnetic path.

Step 4: Cross-section of wire

The conductor area is the RMS current divided by the chosen current density. Pick the standard wire size from the wire table whose copper area is equal to or just above this value.

$$a_w = I_{rms} / J \quad (mm^2)$$

Design flow and checks

The procedure runs in this order for any inductor:

1. Collect L, I_p, I_{rms}, I_{avg} from the converter design, and fix J, B_m and K_w.
2. Compute the required $A_c \times A_w$ in cm^4.
3. Select the E-E core from the catalog with an area product at or above that value, and note its A_c, A_e.
4. Compute the turns N from L, I_p, B_m and A_c, rounded up.
5. Compute the air gap $l_g = \mu_0 N^2 A_c / L$ for that N.
6. Compute the wire area $a_w = I_{rms} / J$ and pick the standard wire size.
7. Recompute the actual flux density $B_m = L I_p / (N A_c)$ with the rounded N and confirm it stays within the material limit.

Worked design: forward converter output inductor

Inductor currents

The inductor carries a ripple current of 0.7 A peak to peak on 1 A average, giving I_p = 1.35 A and I_{rms} = 1.02 A.

Ripple current, with the 1 V diode drop and 1 V ESR drop subtracted from E_s:

$$\Delta I = (E_s - V_d - V_o - I_o R_{esr})\, DT / L = (28 - 1 - 12 - 1) \times 50 \times 10^{-6} / (1 \times 10^{-3}) = 0.7 \text{ A}$$

Parameter	Value
Input voltage V_{in}	24 V ± 10% (design done at nominal 24 V)
Output voltage V_o	12 V
Output current I_o	1 A
Switching frequency f_{sw}	10 kHz (T = 100 µs)
Duty cycle D	0.5
Secondary voltage E_s	28 V
Inductance L	1 mH

Parameter	Value
Inductor ESR	1 Ω (1 V drop at 1 A)
Diode drop	1 V
Flux density B_m	0.25 T (Wb/m²) (ferrite core)
Current density J	3 A/mm² (copper)

Peak, average and RMS current:

$$I_p = I_o + \Delta I/2 = 1 + 0.35 = 1.35\ \text{A}, \quad I_{avg} = I_o = 1\ \text{A}$$

$$I_{rms} = \sqrt{(I_o^2 + (\Delta I)^2/12)} = \sqrt{(1 + 0.49/12)} = 1.02\ \text{A}$$

The inductor current flows for the whole period, so the full-period RMS formula is used. The 0.72 A in the transformer calculation applies to the secondary winding, which conducts only during D.

Why Bm = 0.25 T and J = 3 A/mm²

B_{m} = 0.25 T (ferrite). Ferrite saturates at roughly 0.3 to 0.5 T depending on grade, and the saturation value falls as the core heats up toward 100 °C. Working at 0.25 T keeps the peak flux safely below saturation when the core is hot and when the current rises above nominal, for example at +10% line voltage. A saturated inductor loses its inductance and the current spikes, so the margin matters more than squeezing the core. A moderate B_m also keeps core loss low.

J = 3 A/mm² (copper). Current density sets the copper loss (I^2R) and so the temperature rises of the winding. Around 2 to 4 A/mm² is the usual range for small, naturally cooled magnetics with no fan, and 3 A/mm² sits in the middle: a smaller J needs more copper and window space, a larger J overheats the winding.

Core selection

The required area product is 0.306 cm^4.

Step A: required area product. The inputs, in the units the formula needs:

- I_{rms} = 1.02 A and I_p = 1.35 A
- L = 1 mH = 1×10^{-3} H
- B_m = 0.25 T = 0.25×10^{-4} Wb/cm²
- J = 3 A/mm² = 300 A/cm²
- K_w = 0.6, as used in the transformer calculation on the same record sheet

$$A_c \times A_w = (I_{rms} \cdot I_p \cdot L) / (B_m \cdot J \cdot K_w) = (1.02 \times 1.35 \times 1 \times 10^{-3}) / (0.25 \times 10^{-4} \times 300 \times 0.6) = (1.377 \times 10^{-3}) / (4.5 \times 10^{-3}) = 0.306\ cm^4$$

The catalog lists W_aA_c (window area × core area) in cm^4, so the required value is compared in cm^4 (0.306 cm^4 = 3060 mm^4).

Step B: core selection. The core is selected so that its W_aA_c is equal to or above the required 0.306 cm^4. A core below this value cannot hold the winding and carry the flux together. For example, E 25/10/7 (0_42510EC) has W_aA_c = 0.16 cm^4 and would need about 137 turns, which does not fit in its window.

Selected core: E 25/13/11 (0_42526EC). Catalog data (E, I cores, pages 28 and 29): A_e = 78.4 mm² (A_e min 76.8 mm²), l_e = 57.5 mm, V_e = 4,500 mm³, W_aA_c = 0.41 cm⁴. Since 0.41 cm⁴ is above 0.306 cm⁴, the core is acceptable. The window area is $A_w = W_aA_c / A_e$ = 0.41 / 0.784 ≈ 0.523 cm² (52.3 mm²). The next size down, E 25/13/7 (0_42513EC, 0.27 cm⁴), falls just short.

Size and cost. But increasing the core size increases the cost (more ferrite, a larger bobbin and more board space). So the smallest core with W_aA_c at or above the required value is chosen, not a bigger one.

Number of turns and air gap

The inductor needs 69 turns, and an air gap of about 0.47 mm in the magnetic path to hold L at 1 mH.

With B_m = 0.25 T and A_c = 78.4 mm² = 78.4 × 10^{-6} m²:

$$N = L\, I_p / (B_m\, A_c) = (1 \times 10^{-3} \times 1.35) / (0.25 \times 78.4 \times 10^{-6}) = 68.9 \Rightarrow N = 69 \text{ turns}$$

$$l_g \approx \mu_0\, N^2\, A_c / L = (4\pi \times 10^{-7} \times 69^2 \times 78.4 \times 10^{-6}) / (1 \times 10^{-3}) = (4\pi \times 10^{-7} \times 4761 \times 78.4 \times 10^{-6}) / (1 \times 10^{-3}) = 4.69 \times 10^{-4} \text{ m} \approx 0.47 \text{ mm}$$

Wire size and design summary

The winding needs 0.34 mm² of copper, which is 22 SWG, and the checks all pass.

$$a_w = I_{rms} / J = 1.02 / 3 = 0.34 \text{ mm}^2 \Rightarrow \text{22 SWG (0.711 mm dia., 0.397 mm}^2\text{)}$$

Item	Result
Ripple current ΔI	0.7 A
I_p / I_{rms} / I_{avg}	1.35 A / 1.02 A / 1 A
Required $A_c \times A_w$	0.306 cm⁴
Core chosen	E 25/13/11 (0_42526EC), A_e = 78.4 mm², W_aA_c = 0.41 cm⁴
Turns N	69
Air gap l_g (total)	about 0.47 mm
Wire	22 SWG, 0.397 mm²
Actual current density	1.02 / 0.397 = 2.57 A/mm², below 3 A/mm²
Actual peak flux density $L\, I_p / (N\, A_c)$	0.2496 T, not above 0.25 T
Window fill	69 × 0.397 = 27.4 mm² of copper in a usable window of 0.6 × 52.3 = 31.4 mm²